DEEP LEARNING FOR AUTOMATED AI-GENERATED IMAGE DETECTION

Norbert Hadiprodjo
1010607666, norbert.hadiprodjo@mail.utoronto.ca
Link to Google Colab

Brief Project Description

In today's social media landscape, AI image generation tools now produce highly realistic content at scale, making it difficult to distinguish whether an image is AI-generated or not. This poses significant risks in the spread of disinformation, fraud, and malicious impersonations [1]. Surveys report that nearly half of Canadians encounter deepfakes weekly, and most users want AI-generated content to be clearly labeled to assess credibility [2][3]. Social media companies themselves have acknowledged there is a growing need for automated authenticity detection, introducing detection and labeling initiatives as generative content increases across platforms [4]. This project aims to develop a deep learning–based classifier that determines how likely an image is AI-generated. Deep learning is suited for this task because generative models leave subtle visual and frequency artifacts that are difficult to detect using hand-crafted rules but can be learned by convolutional neural networks [5][6].

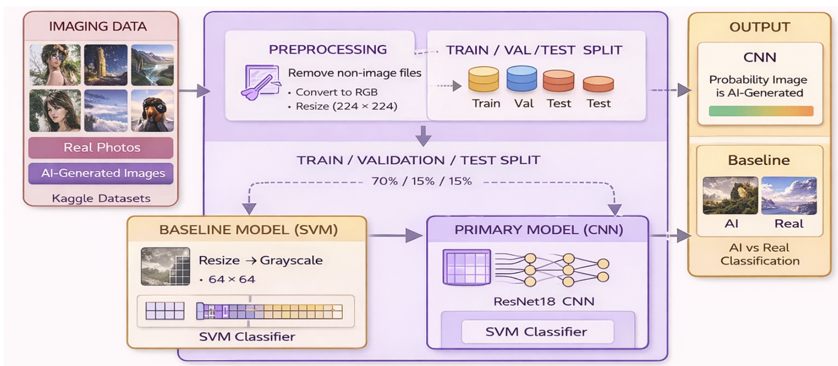

Figure 1: Pipeline for AI-Generated Image Detection

As shown in Figure 1, the overall pipeline for detecting AI-generated images takes input images from Kaggle datasets. The kaggle datasets containing real and AI-generated images are first preprocessed and split into training, validation, and test sets (70/15/15). A baseline SVM model using simple grayscale features and a primary CNN model (ResNet18 with transfer learning) are then trained and compared. The models output a prediction indicating whether an image is real or AI-generated.

Data Processing

The dataset used in this project consists of two classes: real photographs and AI-generated images made from DALL-E and Midjourney, both obtained from AI recognition datasets on Kaggle. The data was organized into two directories corresponding to each class. After filtering the directories to include only valid image formats (e.g., .png, .jpg, .jpeg), the cleaned dataset contained 17,857 AI-generated images and 3,781 real images, for a total of 21,638 images. Non-image files such as metadata files were excluded during dataset loading to prevent errors during training. All images were loaded using the Python Imaging Library and converted to RGB format to standardize the number of color channels across the dataset. After cleaning, all images were resized to 224×224 pixels to match the input requirements of the ResNet architecture used in the primary model. Pixel values were converted to tensors and normalized using PyTorch's ToTensor() transformation. Data augmentation techniques such as random horizontal flipping and color jitter were applied during training to improve model generalization. The dataset was then split into training, validation, and test sets using a 70–15–15 ratio. For final evaluation, real photographs will be collected by the author to ensure the model is tested on previously unseen real-world images, as well as AI-generated images created using other tools like Stable Diffusion, which were not included in the training dataset.

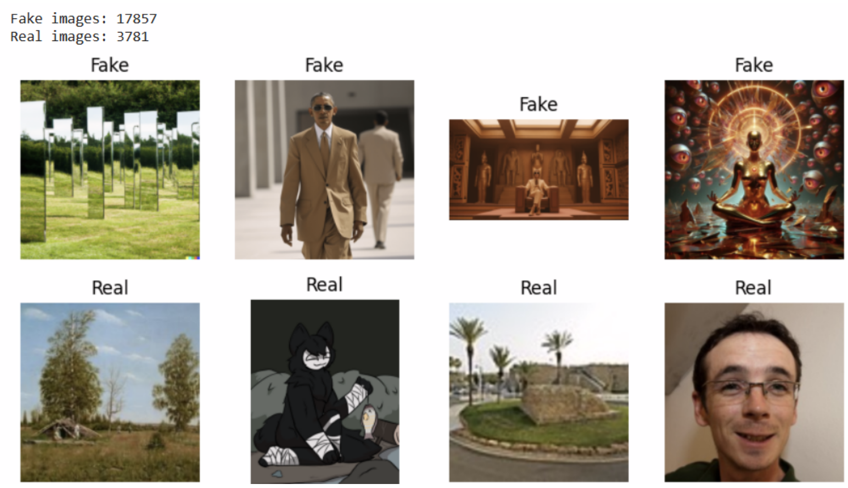

Figure 2: Data Overview

Baseline Model

As shown in Figure 3, a Support Vector Machine (SVM) classifier was implemented as a baseline approach to distinguish between AI-generated images and real photographs. This model uses simple pixel-based features instead of deep learning to establish a reference performance that can be compared against the primary CNN model. Each image was first resized to 128 × 128 pixels and converted to grayscale to reduce computational complexity. The grayscale image was then flattened into a 16,384-dimensional feature vector (128 × 128) representing pixel intensities. These vectors were used as input to the SVM classifier.

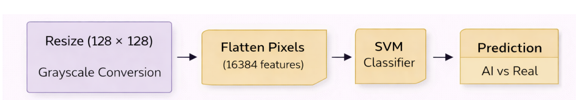

Figure 3: Baseline Pipeline

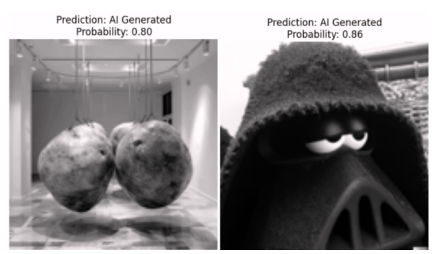

Figure 4: Baseline Prediction

Training this baseline on the AI recognition datasets proved computationally intensive due to the large size of data. Due to the high dimensionality of the features and computational cost of SVM training, a subset of the dataset was used, with the training set containing 4000 images and the testing set containing 1000 images. The SVM model was trained to learn a decision boundary separating real images (label 0) from AI-generated images (label 1). The baseline model achieved an overall classification accuracy of 0.824 on the test dataset. This performance demonstrates that even simple pixel-based features contain some useful information for distinguishing between real and AI-generated images. As shown in Figure 4, the examples illustrate how the SVM classifier correctly identify some AI-generated images. However, because the model relies only on simple pixel-level features, this highlights the limitations of the baseline and motivate the use of a more expressive deep learning model to improve detection performance.

Primary Model

The primary model used in this project is a Convolutional Neural Network (CNN) based on the ResNet18 architecture. A transfer learning approach was used, where a pretrained ResNet18 model from the PyTorch torchvision library served as the backbone of the network. ResNet18 architecture consists of 18 layers, and 11.7 million parameters. For this project, the final fully connected classification layer was replaced with a single output node to perform binary classification between real and AI-generated images. The model uses Binary Cross Entropy with Logits Loss (BCEWithLogitsLoss) as the loss function and is optimized using the Adam optimizer with a learning rate of 1e-4. During preprocessing, input images are resized to 224×224 pixels and converted to tensors. To improve generalization and reduce overfitting, data augmentation techniques including random horizontal flipping and brightness variation using ColorJitter were applied during training. The choice of the ResNet18 architecture is appropriate for this task because CNNs are well-suited for extracting hierarchical spatial features from images. The overall pipeline of this primary model is shown by figure 4 below.

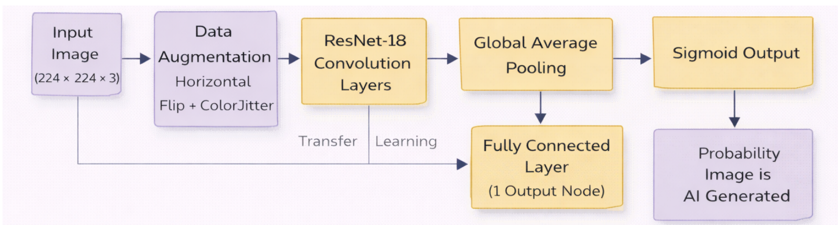

Figure 5: Architecture of the fine-tuned ResNet-18 Model

As shown in Figure 6, the validation accuracy across the five training epochs is approximately 0.97, suggesting that the model generalizes well to unseen data and does not show significant signs of overfitting. Furthermore, figure 7 illustrates qualitative results by showing example predictions from the trained ResNet18 model. The model correctly assigns a low probability of being AI-generated (0.99 close to real) for a natural landscape image, while assigning a very high probability (1.00) for an obviously stylized AI-generated image. This shows that the model can distinguish between realistic photographic textures and artificial visual patterns commonly found in generated images.

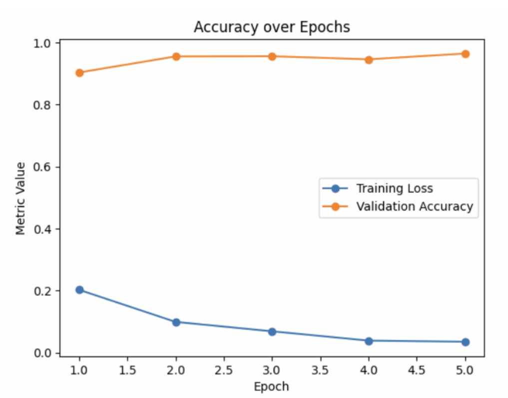

Figure 6: Accuracy curve of primary model.

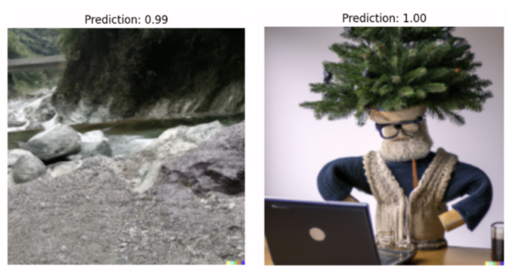

Figure 7: Fine-tuned ResNet-18 model output.

One challenge encountered during the implementation was the large dataset size, which increased the computational requirements. Additionally, some images in the dataset were extremely large, which triggered image loading errors such as the DecompressionBombError in the Python Imaging Library. This issue was addressed by adjusting the maximum allowed image size and resizing all images before training. Overall, these results demonstrate that the ResNet18 architecture provides a strong foundation for detecting AI-generated images, and establishes a solid basis for further improvements and optimization in the final stage of the project.

References

[1] A. Durall, M. Keuper, F.-J. Pfreundt, and J. Keuper, "Unmasking DeepFakes with Simple Features," IEEE International Conference on Image Processing (ICIP), 2020.

[2] A. Singh et al., "Ethical Implications of Generative AI and Deepfake Detection Technologies," Journal of Information Processing Systems, 2024.

[3] Unsplash, "Unsplash: Free High-Resolution Photos," Available: https://unsplash.com

[4] Adobe Inc., "Content Authenticity Initiative (CAI) / Content Credentials," Adobe Research, 2023.

[5] F. Marra, D. Gragnaniello, L. Verdoliva, and G. Poggi, "Detection of GAN-Generated Fake Images Over Social Networks," IEEE Conference on Multimedia Information Processing and Retrieval (MIPR), 2019.

[6] N. Yu, L. S. Davis, and M. Fritz, "Attributing Fake Images to GANs: Learning and Analyzing GAN Fingerprints," IEEE International Conference on Computer Vision (ICCV), 2019.